

Centre Number	Candidate Number	Candidate Name
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NAMIBIA SENIOR SECONDARY CERTIFICATE

AGRICULTURAL SCIENCE ORDINARY LEVEL

6115/1

PAPER 1

2 hours

Marks 120

2024

Additional Material: Non-programmable calculator

INSTRUCTIONS AND INFORMATION TO CANDIDATES

- Write your Centre Number, Candidate Number and Name in the spaces at the top of this page.
- Write in dark blue or black pen.
- You may use a soft pencil for any diagrams, graphs or rough working.
- Do not use correction fluid.
- You may use a non-programmable calculator.
- Do not write in the margin *For Examiner's Use*.
- The number of marks is given in brackets [] at the end of each question or part question.

Section A

- Answer **all** questions.
- Candidates answer on the Question Paper in the spaces provided.

Section B

- Answer any **two** questions.
- Write your answers on the answer sheets at the back of the booklet.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5	
6	
7	
8	
Section B	
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Total	

Marker	
Checker	

This document consists of **20** printed pages.



Republic of Namibia
MINISTRY OF EDUCATION, ARTS AND CULTURE

SECTION A

Answer **all** questions in the spaces provided.

- 1 Fig. 1.1 is a bar graph that shows the number of weeds in seed beds some days after transplanting a cereal crop. One seed bed was covered with mulch while the other seed bed was left without mulch.

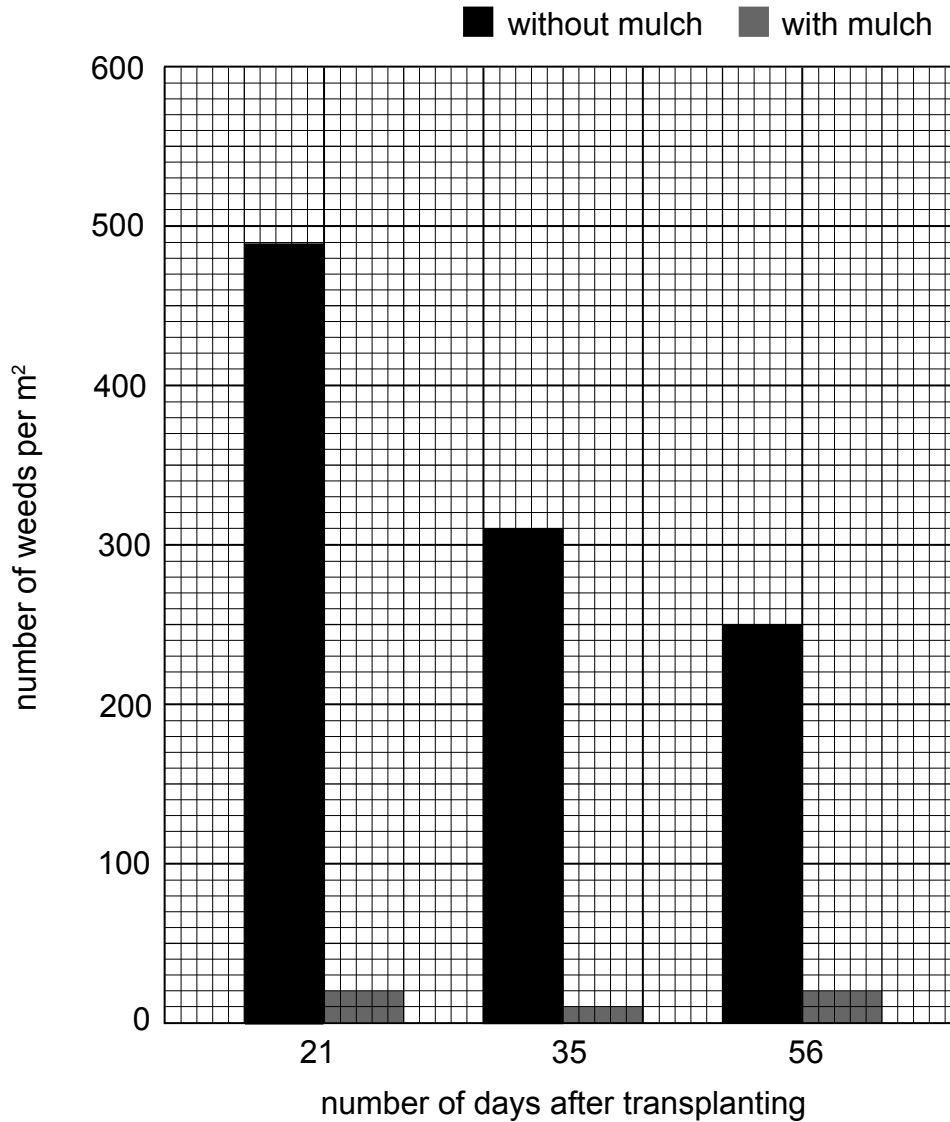


Fig. 1.1

- (a) (i) State the number of days after transplanting when the number of weeds in the seed bed with mulch was the least.
- [1]
- (ii) State the number of weeds 56 days after transplanting in the seed bed.
- with mulch,
- without mulch. [2]

- (iii) Suggest why the number of weeds in the seed bed with mulch was relatively low.

.....

.....

[2]

- (b) Apart from weed control, outline the other ways in which mulching is important.

.....

.....

.....

.....

.....

[3]

- (c) Suggest how low and high humidity might affect a plant negatively.

low humidity.....

.....

high humidity

.....

[2]

[10]

2 (a) State the soil type that fits each of the following descriptions.

1 Sticky when moist, can be molded into any shape and can be polished.

.....

2 Can be molded and sticks to fingers slightly when moist.

.....

3 Feels coarse and is structure less.

.....

4 Has large and small pore spaces, which retain water fairly.

.....

[4]

(b) Fig. 2.1 shows the types of soil erosion **X** and **Y** on agricultural land.

X



Y

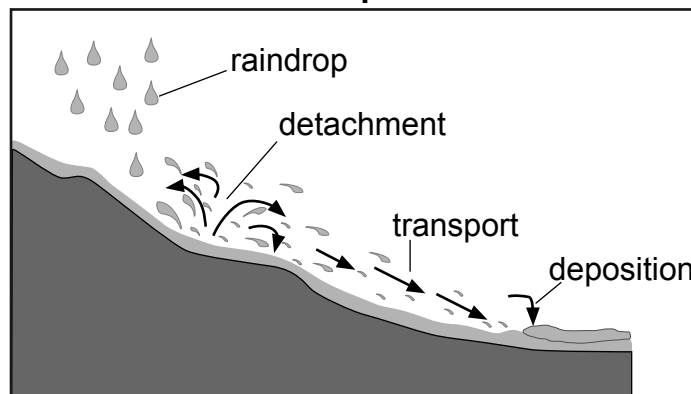


Fig. 2.1

(i) Identify the types of soil erosion, **X** and **Y**, in Fig. 2.1.

X.....

Y.....

[2]

- (ii) Suggest **two** best practices that can be used to restore the land at **X** in Fig. 2.1.

1

2 [2]

- (iii) Explain why zero tillage will be a suitable tilling method on the land at **Y** in Fig. 2.1.

.....

..... [2]

- (c) Fig. 2.2 shows plants growing in undrained and drained soils.

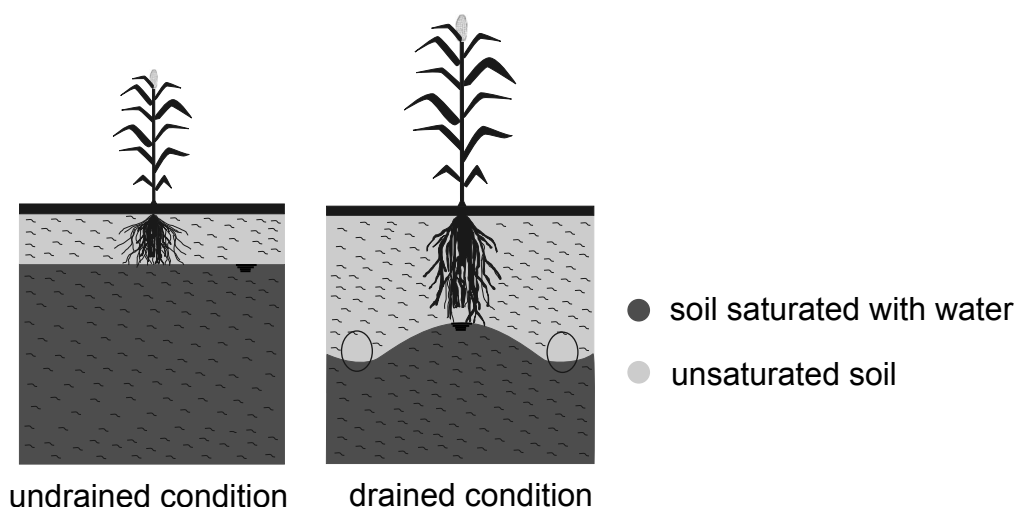


Fig. 2.2

- (i) Identify the drainage system shown in Fig. 2.2.

..... [1]

- (ii) With reference to Fig. 2.2, explain how undrained condition could limit plant growth.

.....

..... [2]

- (d) Why is a waterlogged soil considered unsuitable for crop growth?

- A** has regular supply of water to plants
- B** has optimum condition for seed germination
- C** there is absence of aeration of soil in the root zone of plants
- D** there is a regular supply of nutrients within the root zone of plants

Answer **A, B, C** or **D** [1]

[14]

- 3 Fig. 3.1 shows a plant cell placed in a concentrated sugar solution.

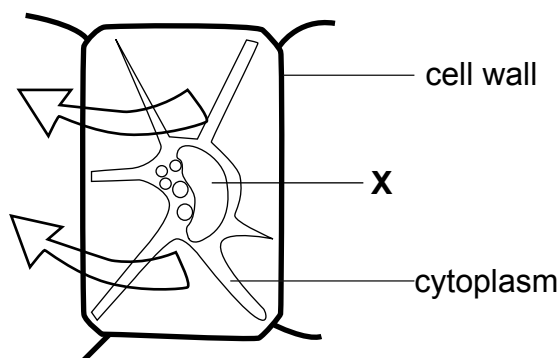


Fig. 3.1

- (a) (i) State what is represented by the arrows on Fig. 3.1.

[1]

- (ii) Describe the process leading to the appearance of the cell shown in Fig. 3.1

[3]

- (iii) Which type of pests feed on the cell sap found in part X?

- A** biting and chewing pests
- B** boring pests
- C** piecing and sucking pests
- D** soil pests

Answer **A, B, C** or **D** [1]

- (b) Describe the passage of dissolved mineral salts from the soil through the vascular tissues.

[3]

[8]

- 4 Fig. 4.1 shows the implement used in land preparation for planting maize.

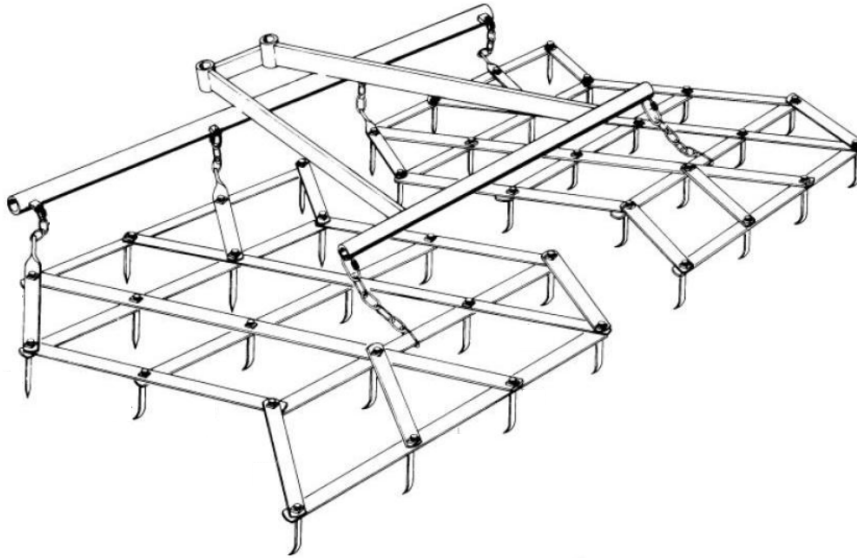


Fig. 4.1

- (a) Identify the implement in Fig. 4.1 and state its function.

name

function

.....

[2]

(b) Fig. 4.2 shows a maize plant.

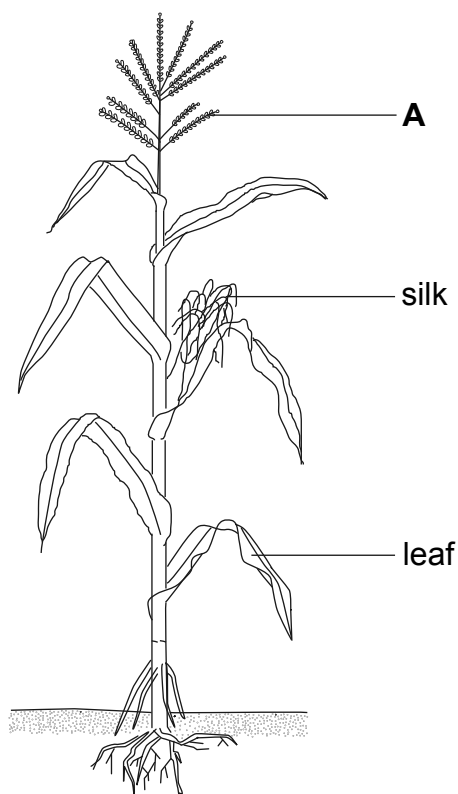


Fig. 4.2

(i) Identify part **A** in Fig. 4.2 and state its function.

name

function..... [2]

(ii) The silk in Fig. 4.2 hangs outside the cob. State the reason why is this an advantage to the plant.

.....

..... [1]

(c) (i) State **two** environmental factors that the plant leaf needs to make sugars.

1

2 [2]

(ii) Suggest the reasons for the following features of a leaf.

large surface area

.....

absence of chloroplasts in the epidermis

..... [2]

[9]

5 (a) Outline **two** functions of protein in animal diet.

1.....

2.....

[2]

(b) Fig. 5.1 shows the influence of the age of pasture on its digestibility, energy levels and protein content.

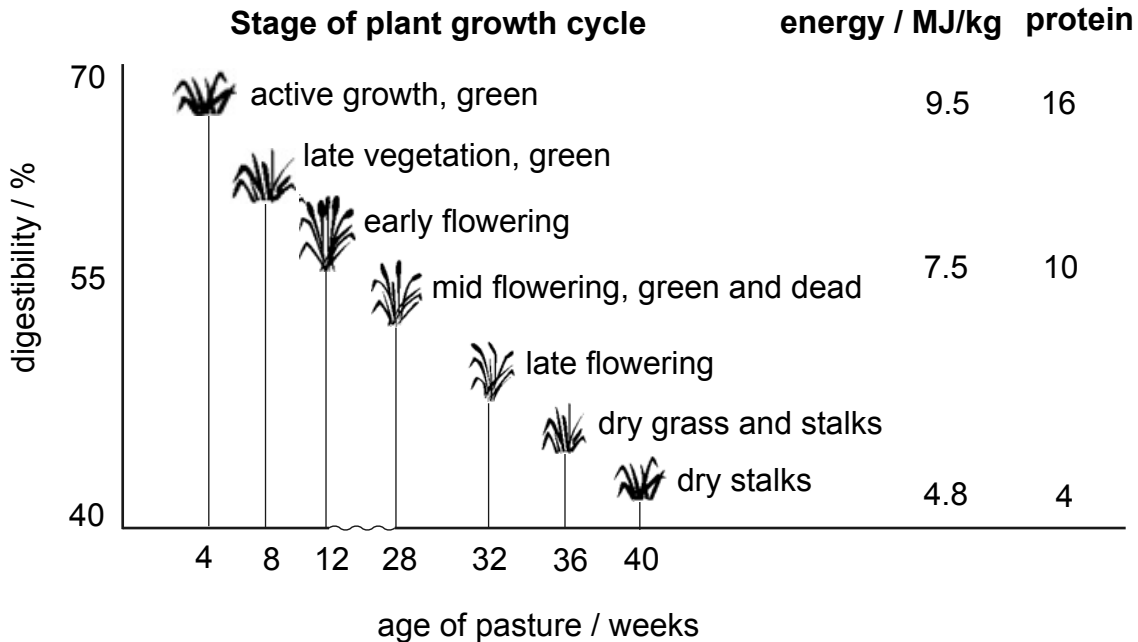


Fig. 5.1

(i) Describe the relationship between the protein content and the age of the pasture.

.....

.....

[1]

(ii) The percentage of digestibility can be used by a farmer as an indicator of when the supplementary feeds can be introduced.

Suggest with a reason, from which percentage of digestibility of the pasture in Fig. 5.1 will the farmer be required to start introducing supplementary feeds.

.....

.....

.....

[2]

- (iii) A sheep eats 10 kg of dry matter that has 40% digestibility.
Work out the dry matter that will be excreted in the faeces.
Show your working and your answer in kg.

[3]

- (c) State **two** disadvantages of a continuous grazing system.

1.....

.....

2.....

.....

[2]

[10]

6 Fig. 6.1 shows the reproductive system of a cow and a bull.

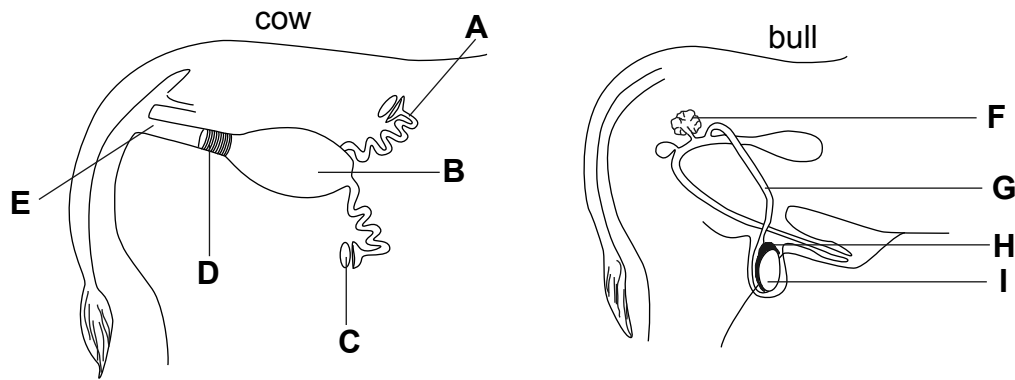


Fig. 6.1

(a) Using letters from Fig. 6.1, indicate in which reproductive part of the farm animal the following events occur.

- 1 semen is deposited during mating.....
- 2 production of fluid that nourishes the sperm cell
- 3 oestrogen is produced
- 4 immature sperm cells are stored

[4]

(b) Fig. 6.2 shows an example of the genetic modification in animals.

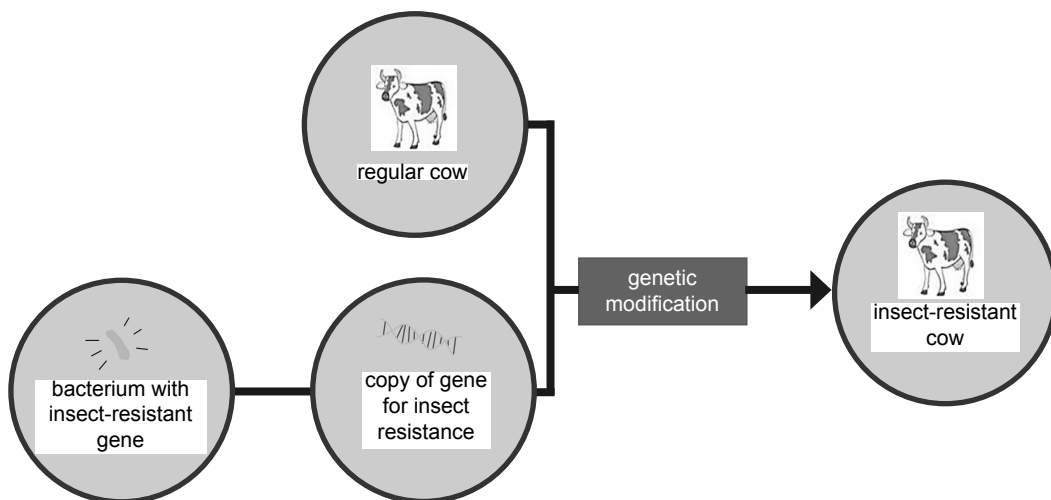


Fig. 6.2

(i) Explain what is meant by the terms *genetic modification* and *gene*.

genetic modification

.....

gene

.....

[3]

- (ii) Other than insect resistance, suggest **three** positive effects of genetic modification to farm animals.

1

.....

2

.....

3

.....

[3]

- (c) Fig. 6.3 shows two beef breeds.

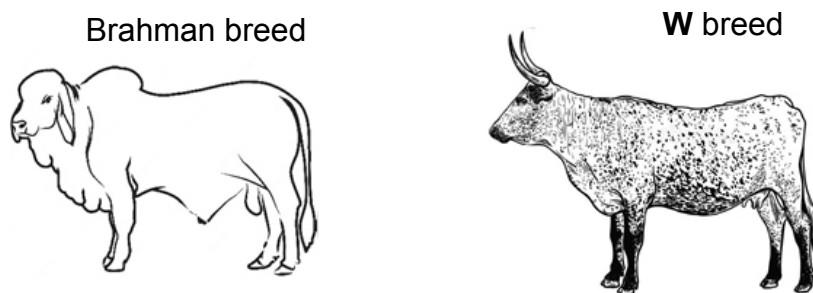


Fig. 6.3

- (i) Identify breed **W**.

.....

[1]

- (ii) Many local commercial farmers prefer to farm with Brahman breed. Outline the characteristics of Brahman that might have influenced their preference.

.....

.....

.....

.....

.....

.....

.....

.....

.....

[4]

- (iii) State **one** threat that breed **W** poses to other animals on the farm.

.....

[1]

[16]

7 Fig. 7.1 shows people cutting down trees in one of Namibia's community forests.

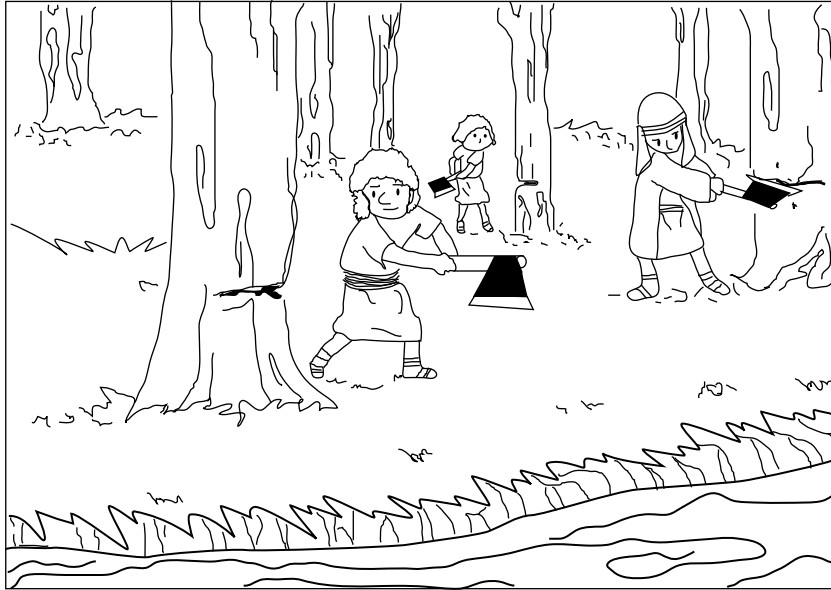


Fig. 7.1

(a) (i) State **two** possible reasons why people cut down trees.

1

.....

2

.....

[2]

(ii) Suggest the problems this community might experience in future as a result of their action in Fig. 7.1.

.....

.....

.....

.....

.....

.....

[4]

- (b) Outline any **four** steps that a community must follow if it wishes to establish a community forest.

1.....

.....

2.....

.....

3.....

.....

4.....

.....

[4]

- (c) Which communal conservancy was among the first four conservancies that were gazetted in Namibia in 1998?

- A** //Gamaseb
- B** Khoadi-//Hôas
- C** King Nehale
- D** Lusesse

Answer **A, B, C** or **D**

[1]

[11]

- 8 Fig. 8.1 shows the output from a maize enterprise.

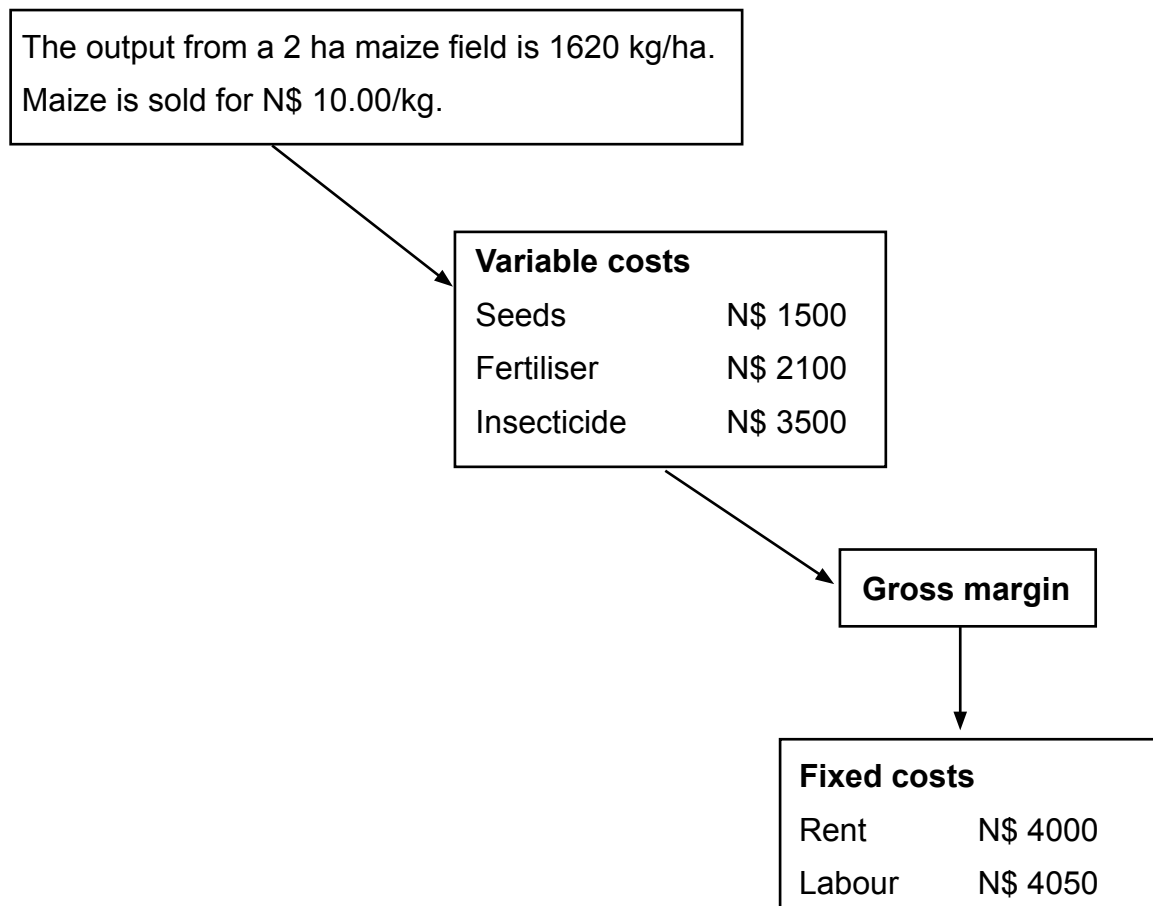


Fig. 8.1

- (a) (i) Define *gross margin*.

.....

[2]

- (ii) Calculate the gross margin obtained from the maize enterprise in Fig. 8.1. Show your working and the formula used.

[4]

- (iii) State whether the enterprise in Fig. 8.1 has made a profit or a loss.
Give a reason for your answer.

.....

.....

.....

[2]

- (b) Fig. 8.2 shows value-added agricultural products **A** and **B**.

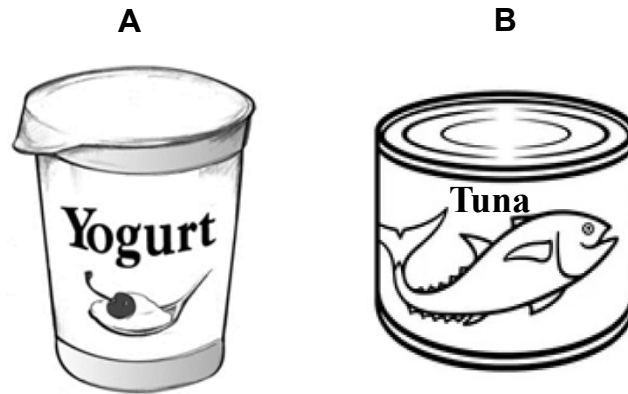


Fig. 8.2

Suggest the methods used in preserving products **A** and **B**.

A

B

[2]

- (c) Explain how the addition of value to agricultural products on the farm may be affected by the occurrence of HIV/AIDS among farm workers.

.....

.....

.....

.....

[2]

[12]

SECTION B

Answer any **two** questions.

Write your answers on the answer sheets provided at the back of the booklet.

- 9** (a) Name **two** countries that import beef cattle from Namibia and explain the international standards that Namibia must meet before beef cattle are exported. [6]
- (b) Discuss five ways in which marketing channels may pose problems in agriculture production. [5]
- (c) Discuss suitable rations that would be required to finish off beef weaners for marketing. [4]
- [15]**
- 10** (a) Name two beef breeds recommended for use in Namibia other than Brahman and describe the importance of their products to the economy. [4]
- (b) Discuss how the breeding cycle of farm animals can be managed in order to increase the yield. [6]
- (c) Explain how keeping a large number of animals than the land can support may lead to a pasture losing its value. [5]
- [15]**
- 11** (a) Discuss the methods that can be employed in preparing the seed bed for planting cereal crops. [5]
- (b) Name **one** leguminous crop and explain its value in maintaining soil fertility. [6]
- (c) Discuss how Integrated Pest Management (IPM) could be used to control pests in crops using cultural methods. [4]
- [15]**
- 12** (a) Describe shifting cultivation and suggest two reasons why it may not be a viable method to Namibian farmers. [7]
- (b) Discuss how Agriculture can help improve the living conditions for rural people in Namibia. [4]
- (c) Explain how the Namibian government can help in the development of rural communities. [4]
- [15]**

[illegible]

The Directorate of National Examinations and Assessment operates under the auspices of the Ministry of Education, Arts and Culture in Namibia.